Notes on lecture 9

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1 Closed and bounded sets in $\mathbb{R}^2$

Definition 1.0.1. Let D be a domain in $\mathbb{R}^2$. A **boundary point** of D is a point (a, b) such that every disk with center (a, b) contains points in D and also points not in D.
The points of D which are not the boundary points are called the **interior points** of D. □

Example 1.0.2. Let $D = \{(x, y) \in \mathbb{R}^2 | x^2 + y^2 \leq 1\}$.
So D is a disk with center $(0, 0)$ and radius 1. The circle $\{(x, y)|x^2 + y^2 = 1\}$ is the boundary of D. □

Figure 1.0.1 illustrates the definition of a boundary point. In Figure 1.0.1, the point (a, b) is a boundary point. In this figure, the thick curve enclosing the region D is the boundary, and all the points lying inside the curve (except the points on the curve itself) constitute the interior.

Definition 1.0.3. A **closed set** in $\mathbb{R}^2$ is a set that contains all its boundary points. □

Remark 1.0.4. If even one point on the boundary is omitted, the set would not be closed.

Figure 1.0.2 explains the concept of a closed set. In part (a) of this figure, we can see some closed sets, which contain all their boundary points. And in part (b) of this figure, we can see some sets which are not closed, because they do not contain all their boundary points.

Definition 1.0.5. A **bounded set** in $\mathbb{R}^2$ is a set which is contained in a disk in $\mathbb{R}^2$. That is, the extent of that set is finite. □

Example 1.0.6. Let $D_1 = \{(x, y) \in \mathbb{R}^2 | x = y\}$. Clearly, D_1 cannot be conatined in any disk in $\mathbb{R}^2$. So D_1 is NOT a bounded set. □

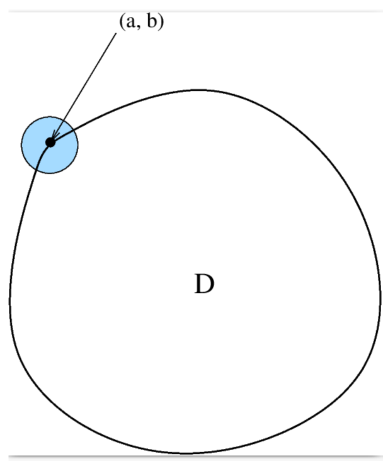

Figure 1.0.1: Boundary point

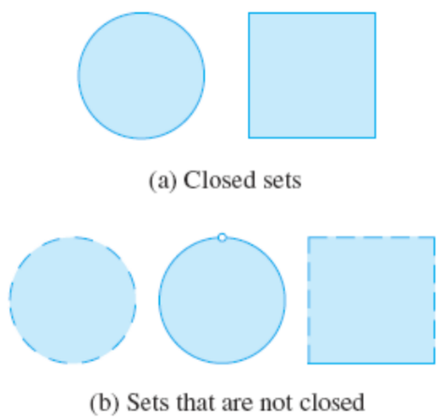

Figure 1.0.2: Closed sets and not closed sets

2 Extreme value theorem for a function of 2 variables

Theorem 2.0.1. *Let* $f : D(\subseteq \mathbb{R}^2) \to \mathbb{R}$ *be a function of 2 variables, where* D *is a closed and bounded domain in* $\mathbb{R}^2$*. Let* f *be continuous. Then* f *posseses its absolute maximum value* $f(x_1, y_1)$ *at* (x_1, y_1)*, and its absolute minimum value* $f(x_2, y_2)$ *at* (x_2, y_2)*, where* (x_1, y_1) *and* (x_2, y_2) *belong to* D*.*

To find the absolute maximum and minimum values of a continuous function f on a closed, bounded set D, we should take the following steps:

1. Find the values of f at the critical points of f in D.
2. Find the extreme values of f on the boundary of D.
3. The largest of the values from steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.

Example 2.0.2. Find the absolute maximum and minimum values of f on the set D, where $f(x,y) = x^2+y^2-2x$, and D is the closed triangular region with vertices $(2,0), (0,2)$, and $(0,-2)$.
Let B_1 denote the boundary of D which is the straight line connecting the points $(0,-2)$ and $(2,0)$. Let B_2 denote the boundary of D which is the straight line connecting the points $(2,0)$ and $(0,2)$. And let B_3 denote the remaining boundary of the triangular region D.
We first need to find the critical points of f in D. To find the critical points, we should equate f_x and f_y to zero.

$$f_x = 2x - 2 = 0 \Rightarrow x = 1$$

and

$$f_y = 2y = 0 \Rightarrow y = 0.$$

So the only critical point is $(1,0)$ and $f(1,0) = -1$.
Clearly, the equation of the straight line B_1 is $y = x-2$ (where $0 \le x \le 2$), and that of the straight line B_2 is $y = 2-x$ (where $0 \le x \le 2$).
Consider B_1 first. On B_1, we have $f(x,y) = f(x,x-2) = x^2+(x-2)^2-2x = 2x^2-6x+4$. Now to find the maximum or minimum value(s) of f on B_1, we can apply the extreme value theorem for functions of one variable. But we can also find it out in some other way as follows: Observe that

$$2x^2 - 6x + 4 = 2(x - \frac{3}{2})^2 - \frac{1}{2}.$$

So for minimum value, $x = \frac{3}{2}$ and $y = \frac{3}{2} - 2 = -\frac{1}{2}$ and $f(\frac{3}{2}, -\frac{1}{2}) = -\frac{1}{2}$. And the maximum value on B_1 is going to occur at $x = 0, y = -2$ and $f(0,-2) = 4$.
Let us consider B_2 now. As in the case of B_1, the minimum value of f on

B_2 will be attained at the point $(\frac{3}{2}, \frac{1}{2})$ and the minimum value of f (at this point) is $-\frac{1}{2}$. Also, the maximum value of f on B_2 will be attained at the point $(0, 2)$ and the maximum value of f (at this point) is 4.
Along B_3, we have $x = 0$ and $-2 \leq y \leq 2$. The maximum value of f on B_3 will be attained at the points $(0, \pm 2)$ and the maximum value at these 2 points is 4. Similarly, the minimum value on B_3 is attained at the point $(0, 0)$ and the minimum value is 0.
So among all these, the absolute maximum value of f is 4 and the absolute minimum value is -1. The points where these values are attained are $(0, \pm 2)$ and $(1, 0)$ respectively. $\square$

3 Extreme values by Lagrange multipliers

Let us start this section with the following motivating example:

Example 3.0.1. A rectangular box without a lid is to be made from 12 square metres of cardboard. Find the maximum volume of such a box.
If we consider the length to be x metres, breadth to be y metres, and height to be z metres, then the volume $V(x, y, z) = xyz$ cubic metres. And the surface area equals $xy + 2xz + 2yz$, which in turn equals 12 because the box is to be made from 12 square metres of cardboard. Let

$$g(x, y, z) = xy + 2xz + 2yz.$$

The problem now becomes: Find the maximum value of $V(x, y, z)$ subject to the condition $g(x, y, z) = 12$.
Now $g(x, y, z) = 12 \Rightarrow z = \frac{12-xy}{2(x+y)}$. Note here that we have $2(x + y)$ in the denominator. But since x and y are the length and breadth respectively of the box, they are always > 0. So the denominator never vanishes. Putting $z = \frac{12-xy}{2(x+y)}$ in the formula for $V(x, y, z)$, we get

$$V(x, y, z) = \phi(x, y) = \frac{xy(12 - xy)}{2(x + y)}.$$

In this way, we can now visualize V as a function ϕ of 2 variables x and y. Now all that we need to do is to find out the maximum value of $\phi(x, y)$. The remaining work is a homework for you. $\square$

Earlier, when we were calculating the maximum or minimum value(s) of a function, there were no constraints. But now, in the method of Lagrange multipliers, we have to maximize/minimize a function subject to some constraint(s).

Suppose f is a function of 2 variables x and y and we have to maximize/minimize the value of f subject to the condition $g(x, y) = k$. Clearly $g(x, y) = k$ represents a curve in the xy-plane. We are seeking the extreme values of $f(x, y)$ when the point (x, y) is restricted to lie on the curve $g(x, y) = k$. For convenience, let us try to see the problem of maximizing $f(x, y)$ subject to $g(x, y) = k$.

Let us draw the level curves corresponding to the surface $z = f(x, y)$. To maximize $f(x, y)$ subject to $g(x, y) = k$ is equivalent to finding the largest value of c such that the level curve $f(x, y) = c$ intersects $g(x, y) = k$. Figure 3.0.1 explains this phenomenon.

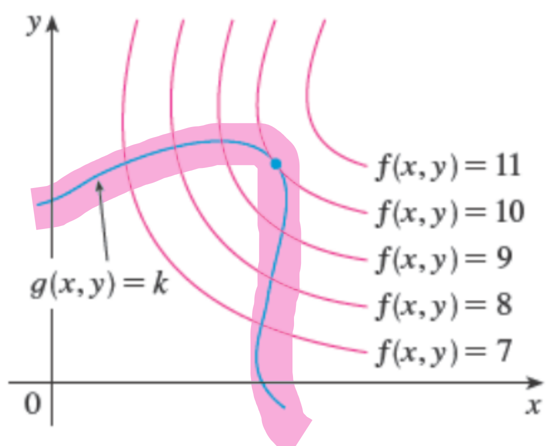

Figure 3.0.1: Maximizing $f(x, y)$ subject to $g(x, y) = k$

It appears from Figure 3.0.1 that this happens when these curves just touch each other, that is, when they have a common tangent line (Otherwise, the value of c could be increased further). This means that the normal lines at the point (x_0, y_0) where they touch each other, are identical. So the gradient vectors are parallel, that is,

$$\nabla f(x_0, y_0) = \lambda \nabla g(x_0, y_0) \text{ for some scalar } \lambda \neq 0.$$

The scalar λ is called a **Lagrange multiplier**.

Remark 3.0.2. We are taking the scalar λ to be nonzero because if $\lambda = 0$, then the normal will be the zero vector and we will not get any direction in that case.

Example 3.0.3. Find the points on the rectangular hyperbola $xy = 1$ that are closest to the origin $(0, 0)$.
Take any point (x, y) in the xy-plane. Now distance of (x, y) from the origin is $\sqrt{x^2 + y^2}$.
We are trying to minimize $\sqrt{x^2 + y^2}$ subject to $xy = 1$. Now minimizing $\sqrt{x^2 + y^2}$ is equivalent to minimizing $f(x, y) = x^2 + y^2$.
Here $g(x, y) = xy - 1$. Now we will convert f into a function $\phi(x)$ of the single variable x with the help of the constraint $g(x, y) = 0$. Since $g(x, y) = 0 \Rightarrow xy = 1$, we have

$$f(x, y) = \phi(x) = x^2 + \frac{1}{x^2} = \frac{1 + x^4}{x^2}.$$

We need to equate $\phi'(x)$ to 0 for finding the point where the minimimum is attained. After equating $\phi'(x)$ to zero, we will get $x = \pm 1$. Since $xy = 1$, we get $y = \frac{1}{x} = \pm 1$. So the points on the rectangular hyperbola which are closest to the origin are $(1, 1)$ and $(-1, -1)$.

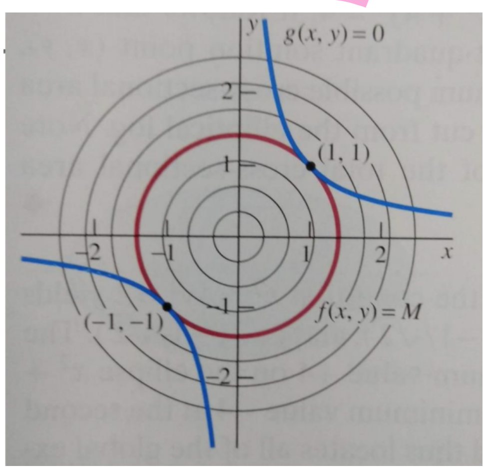

Figure 3.0.2: Figure for Example 3.0.3

Observe in Figure 3.0.2 that the blue curve (the constraint $xy = 1$) and the red closed curve (the level curve of the surface $z = f(x, y) = x^2 + y^2$) are just touching each other at the points $(1, 1)$ and $(-1, -1)$. $\square$

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